

NeuroMotion: End-to-End Notes

From Movement to Synthetic EMG

Project Notes

May 29, 2026

1 What this repository does

Goal: generate realistic multi-channel synthetic EMG from a planned hand/wrist movement.

The repository links four pieces:

1. Musculoskeletal simulation (OpenSim model, joint kinematics, muscle lengths)
2. Motor neuron pool physiology (recruitment and spike trains)
3. MUAP waveform generation (BioMime model)
4. Signal composition (sum MU contributions across channels over time)

2 Main folders and their role

Path	Purpose
NeuroMotion/MSKlib	OpenSim movement + muscle-parameter extraction
NeuroMotion/MNPoollib	Motor neuron pool model + spike train generation
NeuroMotion/EMGSyn.py	EMG synthesis logic and orchestration
scripts/mov2emg.py	End-to-end runnable pipeline
ckp/	Config and pretrained weights (BioMime model)

3 What is the .osim model and why it matters

What .osim is

A .osim file is the OpenSim model definition: bones, joints, constraints, muscles, tendon paths, and mechanical properties.

Why it is needed here

Without .osim, movement is only a list of angles. With .osim, the code can compute muscle-specific biomechanical changes over time, for example fiber length trajectories. Those trajectories condition neural and MUAP behavior, improving physiological realism.

Difference between `.osim` and `.mot`

- `.osim`: model specification (anatomy and mechanics)
- `.mot`: time-series motion (joint coordinates over time)

4 Pipeline from beginning to end

Step 1 — Define movement trajectory

In `scripts/mov2emg.py`, a movement sequence is defined as poses and durations. This is converted into a smooth joint-angle trajectory sampled in time.

Step 2 — Run OpenSim and extract muscle changes

`MSKModel` applies each joint state to the OpenSim model and computes muscle lengths. Then `len2params()` derives parameter changes (e.g., depth/CV/length trends) used downstream.

Step 3 — Simulate motor neuron pool activity

For a target muscle (e.g., FDSI), `MotoneuronPool` initializes motor units and produces spike trains from neural drive/excitation.

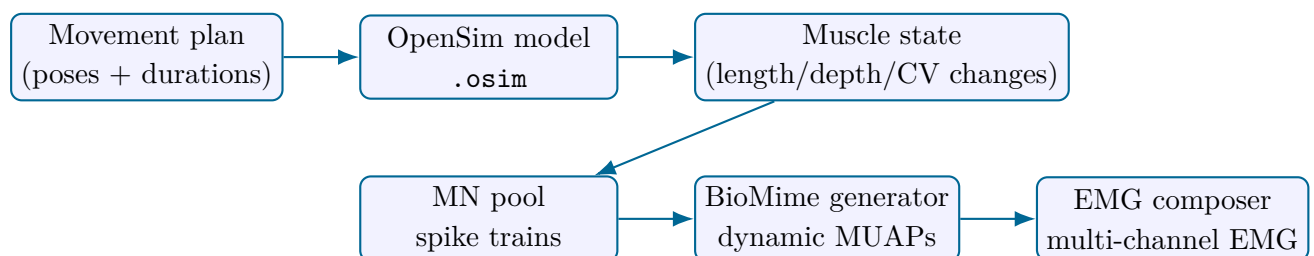
Step 4 — Generate dynamic MUAPs (BioMime)

The generator model takes physiological conditioning variables and produces MUAP waveforms per motor unit and per time step.

Step 5 — Compose final EMG signal

EMG channels are synthesized by summing all active MU contributions at spike times. The script saves figures (spikes, MUAP snapshots, EMG image) and output arrays.

5 Visual overview



6 How EMG is generated (mathematical view)

For each channel c , synthetic EMG is formed by superposition:

$$\text{EMG}_c(t) = \sum_{m=1}^{N_{MU}} \sum_{k \in \mathcal{S}_m} \text{MUAP}_{c,m}(t - t_k) + n_c(t)$$

where:

- $\mathcal{S}_m$ = spike times of motor unit m
- $\text{MUAP}_{c,m}(\cdot)$ = channel-specific waveform for unit m
- $n_c(t)$ = optional noise/perturbation term

7 Practical run path in this repo

- Main run script: `scripts/mov2emg.py`
- Movement visualization utility: `scripts/visualize_movement.py`
- OpenSim movement export: `MSKModel.write_mov(...)` creates `.mot`

Typical generated outputs include:

- `figs/spikes_*.jpg`
- `res/emg_*.jpg`
- `res/*muap*.jpg`
- `vis_output/movement.mot`

8 Why this pipeline is useful

- Create labeled EMG data without repeated human data collection
- Stress-test decomposition and gesture-recognition algorithms
- Study parameter effects (recruitment, MUAP morphology, movement dynamics)
- Prototype quickly before wet-lab/hardware experiments

9 Quick compile command

From repository root:

```
pdflatex -output-directory notes notes/neuromotion_notes.tex
```

If `pdflatex` is not installed:

```
sudo apt update  
sudo apt install -y texlive-latex-extra texlive-fonts-recommended
```